

Harness Lighting System

Week 5 / Complete Report / F2

Peter | 2026-09-29 | Computational study · Preliminary engineering

F2: complete Week 5 computational handover

29 September 2026. This edition supersedes the current conclusions and requirement mapping in F1 using led-reqs-detailed.md. F1 and all original daily evidence are preserved. This is a completed update of the computational delivery with available inputs, not product design release or physical acceptance.

The revised delivery adds R1–R11 / M1–M8 tracking, eight conditional window/layout scenarios, a new 12-source dynamic optical basis, all listed state families, A/B regimes, current budgets, and Windows/macOS offline launch packages. Unknown physical quantities remain unknown.

Section	Page
F2 overview and contents	1
Firm, provisional and unknown inputs	2
Geometry and Blender / Workbench models	3
Supplementary Speos results	4
Numerical checks and optical limits	5
R1–R5 requirement coverage	6
R6–R11 requirement coverage	7
Regimes, state language and temporal model	8
Electrical and thermal budget	9
Camera, PWM and optical isolation	10
Web App, playback and portable delivery	11
Open evidence, handover and next decisions	12
A / Day 1: Geometry audit	13–15
B / Day 2: Static optical baseline	16–18
C / Day 3: Material/layout sensitivity	19–22
D / Day 4: Dynamics and exposure	23–27
E / Day 5: Original R1–R7 draft	28–33

02 Input status: avoid converting assumptions into specifications

Status	Input and consequence
Specified in supplied update	SK9822 / APA102-class RGB, two-wire interface, 24-bit RGB plus 5-bit global brightness; exact supplier part, flux/current curves and PWM implementation remain unknown.
Specified floor	At least 12 evenly spaced emitters. Vendor samples with 8–9 do not meet that stated count floor; 24 is a study candidate, not an adopted design.
Specified optical requirements	At least 100 lx added on axis at 0.15 m; window uniformity within $\pm 20\%$. The 90-degree coverage wording and uniformity reference/protocol need clarification.
Open / provisional	Window OD45 versus OD52 remains open. Hole $\varnothing 14$ is specified. Gap about 10 mm, with uniformity from about 8 mm, is provisional.
Calculation assumption	60% diffuser transmission is not measured. The supplied 240mA illuminance table is preserved as a calculated baseline; it is not used to calibrate the new solver model.
Specified camera / scene	IMX415 rolling shutter, f/2, fixed focus about 0.6m, $82^\circ \times 52^\circ$ FOV, fixed IR-cut; typical rooms 100–500lx, subjects 0.3–1.5m; no ambient light sensor.
Budgets / policy	200mA sustained, up to 500mA bursts, 500mA absolute max. About 7W for the sealed metal sphere. Existing slew policy blanks LEDs to protect the 5V rail.
State policy	White normally; red reserved for camera live; asleep is dark with camera off. Optional white emitters, IR, larger connector or work light are not approved changes.

Only the supplied detailed requirements were available. Referenced supplier documents and the Investigation brief tab were not provided; their contents have not been independently verified.

03 Conditional geometry study and saved engineering models

Eight cases combine window OD45/52, 12/24 evenly spaced emitters and an 8/10mm gap. To make a controlled comparison without final front-stack CAD, the study uses a planar annular window, hole Ø14, assumed 2mm thickness and 2×2mm Lambertian emitting faces. LED centre radius is the midpoint of the annular radial span: 14.75mm for OD45, 16.5mm for OD52. The gap is from emitter plane to the inner window face.

These are envelope assumptions, not a replacement production CAD. The historical curved CAD remains preserved. Do not rescale that shell to OD45/52 or treat its former mounting dimensions as confirmed by the new document.

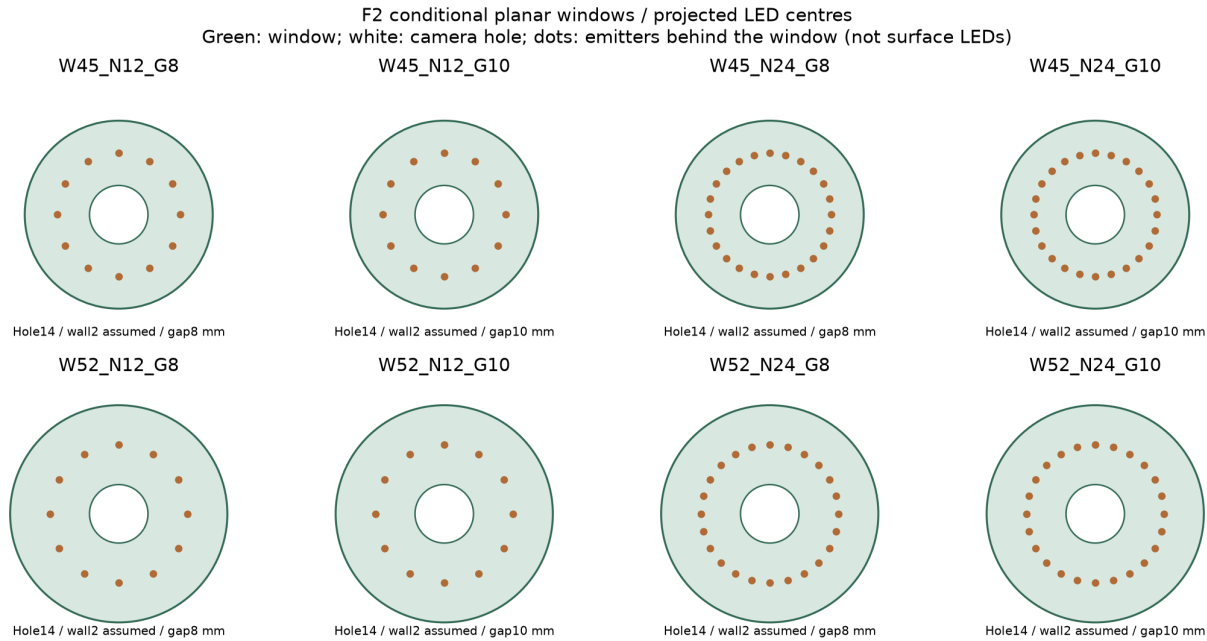


Figure F2-1. Eight configurations matching the Blender study. Dots project emitter centres behind the window; colours are diagram labels, not optical materials.

Artifact	Meaning
Harness_Week5_F2_Envelope_Study.blend	Native Blender scene; original scene retained; 8 labeled configurations and LED placeholders.
Conditional_Window_OD45/52_Hole14_T2.step	Valid annular STEP solids, in mm; provisional flat-window geometry only.
Harness_Week5_F2_Geometry.wbpj	Workbench preparation with OD45 conditional STEP assigned; actual optical solves are the separate Speos RPC exports.

04 Supplementary Speos optical results

Completed 22 successful solves: eight 2M-ray static screens, an 8M-ray reference, twelve 2M-ray single-source responses, and an 8M-ray unequal-output check. All use 555nm, 1lm total source flux for static comparisons, $n=1.5$ and hypothetical HG scattering 1/mm, $g=0.7$. Distances 0.15/0.3/0.5/1/1.5m are measured from the window front, unlike the original source-plane reference.

Case	Axis patch lx/lm	Mean at 0.5m lx/lm
W45_N12_G10	6.410	0.594
W45_N12_G8	6.745	0.635
W45_N24_G10	6.370	0.588
W45_N24_G8	6.885	0.638
W52_N12_G10	7.325	0.641
W52_N12_G8	7.590	0.680
W52_N24_G10	7.040	0.636
W52_N24_G8	7.280	0.676

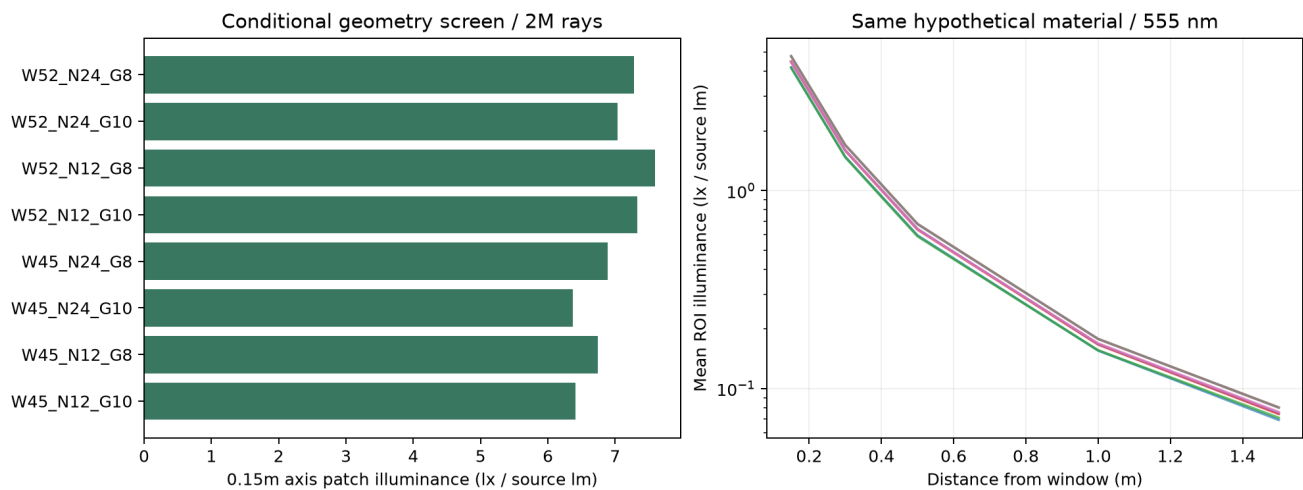


Figure F2-2. Conditional geometry sensitivity, not actual RGB hardware output.

The W45_N12_G10 per-source basis gives 6.472 lx per emitted source lumen averaged over the 10×10mm axial patch at 0.15m. It requires about 15.45 lm at all-on command to obtain 100lx in that patch under these assumptions. This is not a point-axis measurement or proof that the real ring can supply that flux within 200mA.

05 Verification, uniformity and limits

Check	Result / scope
Direct-versus-superposition means	Maximum relative mean difference 0.56%; criterion 3%.
Direct-versus-superposition maps	Maximum RMS/reference mean 4.30%; criterion 6%, with maximum-bin error/reference mean limited to 20%.
Axial patch comparison	Maximum difference 0.63%; criterion 6%. Small detector area is more sensitive to ray noise.
Energy and propagation	Every accepted solve has zero propagation errors and zero max-impact stops; photometric energy closure checked within 1e-4lm.
Spatial interpretation	Live maps are 5×5 bins over a 200mm square, each bin 40mm. Axis diagnostic is a 10mm square. Cone diagnostic uses 10mm pixels inside a projected ±45° cone at 0.15m. None establishes pointwise or cone-wide 100lx acceptance.

The above tolerances are computational screening criteria, not product tolerances. Static 2M/8M comparison is recorded in summary.json. Single-source superposition is checked with equal and unequal outputs; the seven other geometries have static evidence only.

For R10, the provisional statistic is maximum absolute deviation from the sampled window mean. Pixels crossing the aperture or outer boundary are excluded. The 2M and 8M front-view radiance maps contain significant fine-pixel noise; whole-window/all-angle convergence is incomplete. No candidate is declared compliant with ±20%, and noisy extrema are not presented as measured product failure.

The new basis does not represent RGB-mixed white or capture red. Actual spectra, angular intensity, packaging, diffuser BSDF/transmission, PCB/housing occlusion, camera lens and source-to-current calibration are absent. The specified 60% transmission was not silently assigned to the hypothetical material.

06 Requirements R1–R5: revised coverage

ID / Test	Requirement and current evidence
R1 / M1	≥ 100 lx added at 0.15 m on axis; CV distance from M1 0.15m axis-patch and cone sensitivity solves; real flux and CV not known
R2 / M1	No full-off animation in A; floor level from M1 A waveform baseline implemented; hardware PWM and slew conflict unresolved
R3 / M1	Camera exposure/gain minus calibrated ring contribution; no sampling blink; awake only Algorithm contract and simulator-only ambient input; no calibrated detector
R4 / M3;M4	14 state families; white only; red reserved for camera live Proposed state simulator covers all listed states, not approved UX
R5 / M5	Captured colour/level change within M5 tolerance Red capture policy shown; no RGB spectral/image simulation or passing claim

Regime interpretation

Regime A uses the ring as scene light in dim conditions or close range; Regime B treats it primarily as a signal in typical lit rooms. The actual boundary and CV floor remain M1 measurements. Distance alone does not classify a real scene. Manual app selection is a scenario control, not an implemented darkness detector.

The no-ALS algorithm requires calibrated camera exposure/gain/image statistics and a calibrated ring contribution in the same measurement space, with hysteresis. Simply subtracting a target-plane lux value from camera gain is invalid. The app does not blink to sample ambient light, and makes no asleep estimate.

07 Requirements R6–R11: revised coverage

ID / Test	Requirement and current evidence
R6 / M3;M4	≤3 flashes/s policy; smooth fades; depth/time from M3/M4 Proposed slow cadences and transitions; no perceptual/WCAG certification
R7 / M6	No visible bands over actual exposure range; vendor PWM/global-brightness disclosure Ideal PWM calculator retained; actual second-PWM and sensor timing unknown
R8 / M3	Frame brightness within M3 limit or provide ring level to image processor Exported per-frame commands as integration interface; AE not modeled
R9 / M7	Contrast loss through Ø14 aperture within M7 tolerance Hole modeled; no lens/baffle/contrast solve; opening flux is not flare
R10 / M2	≤±20%; no hotspots/dark sectors; measured material from M2 Conditional sampled mean-relative deviation reported; full window/views unverified
R11 / M8	200mA sustained; 500mA burst/absolute max; 7W sphere envelope Electrical budget calculations and calibrated-input estimator; real current/thermal pending

Traceability: former R1/R2 retain their principal roles; former R3 becomes R4, former R4 becomes R6, former R5 becomes R10, former R6 expands into R7/R8, former R7 becomes R11. R3/R5/R9 are separately added concerns. Historical appendix IDs must be interpreted using this mapping.

08 Regimes and complete proposed state language

State family	Proposed simulator behaviour; not product approval
Booting / waking	Smooth rise. Waking begins from darkness; the ramp does not declare camera/CV ready.
Idle / passive	Low steady white; A adds its assumed baseline, B does not enforce one.
Listening / speaking	Smooth pulse / opposed-lobe modulation around the A baseline. Speech envelope is synthetic.
Capturing	Solid red policy is shown; equal-flux monochromatic lux is only a proxy. M5 colour/level compatibility unresolved.
Capture complete	One smooth brightness decay, replacing the unqualified bright-flash concept in this proposal.
Slewing	Blank according to existing rail policy. Explicit conflict in A; not silently overridden with a floor.
Volume / battery / OTA	Contiguous gauge/progress arc over 12 channels. A keeps all channels at least at baseline; B permits dark inactive segments.
Error	Smooth pulse with configurable period at least 0.5s; no proposed fast full-off error blink.
Asleep	All optical commands zero, camera off; dark-command electronics may still consume standby current.

The full sequence visits all 14 state families for at least 3s each: default 42s, 84,001 samples at 0.5ms including both endpoints. Fade duration and pulse period change the physical waveform; playback speed changes viewing only. Normal white-to-red policy changes and sleep/slew interruptions remain explicit.

Continuous command floors do not guarantee instantaneous optical floors under full PWM. Low-level RGB quantization, true carrier timing and human comfort remain unverified. A rate below three flashes per second is not, by itself, a full accessibility or photosensitivity assessment.

09 Electrical and thermal budgets

Budget calculation	Consequence
5V × 200mA = 1.0W	Continuous ring input budget; does not prove available optical flux.
5V × 500mA = 2.5W	Burst and absolute connector envelope; no approved burst duration.
12 emitters: 16.67 / 41.67mA each	Equal-share continuous / burst ceilings before logic/standby overhead, summed over RGB channels; not a channel-current rating.
24 emitters: 8.33 / 20.83mA each	More emitters divide the same total budget; equal-flux optical screening is not equal-current testing.
About 7W sphere envelope	Whole-system budget, not power reserved for LEDs. The 0.2W four-LED line is stale.
Slew rail policy	5V rail kept at ≤2.9A of 3A by blanking LEDs. A floor during slew requires a system power-policy decision.

The web estimator requires full-command and dark-command ring currents from the same calibrated operating point as the entered optical flux: $I(t) = I_{\text{standby}} + (I_{\text{full}} - I_{\text{standby}}) \text{mean}(\text{command})$. Unknown currents remain blank. RGB colour and carrier spikes can invalidate one common curve; the estimate cannot certify the instantaneous 500mA limit.

Steady single-node rise is optionally estimated as $\Delta T = R_{\text{theta}} \times 5V \times I_{\text{mean}}$. It does not predict junction temperature, local hotspots, sphere temperature distribution or transient/burst duration. A meaningful 3D thermal solve still needs real losses, PCB/metal geometry, materials, contacts, ambient conditions and limits. No fabricated thermal model has been presented as hardware evidence.

The provided 240mA calculated illumination baseline is above the 200mA sustained budget. Original values are preserved and not extrapolated into a new contracted rating. M8 current/flux and thermal measurements are needed to connect it to sustained operation.

10 Camera, PWM and optical isolation

Known / modeled	Remaining condition
IMX415 / rolling shutter / fixed IR-cut	Exact sensor mode, frame/line timings, exposure/gain limits and spectral response are not provided. IR illumination is not a software-only option.
Existing 100/120Hz anti-flicker	Does not establish immunity to LED PWM or a second global-brightness PWM. Vendor disclosure and M6 image testing remain necessary.
Ideal PWM calculator	Exact all-phase bounds for one ideal rectangular carrier and constant duty. No dual-carrier driver, actual IMX415 image, rolling-row/AE closed-loop or physical passing claim.
R8 exposure stability	Exports include per-time mean command for a possible future ISP feedforward interface. No current connection to an ISP or validated AE controller.
R5 red capture	Colour shift depends on real RGB spectrum, scene reflectance and camera spectral response; equal photometric output is not equal image colour or sensor response.
R9 hole / baffle	Ø14 aperture is present in geometry. No real lens or light seal is specified; target maps and historical opening flux cannot establish image contrast loss.

The previous Day4 rolling-exposure cases remain an explicitly hypothetical historical sensitivity study. Their 480 virtual rows and assumed row time are not relabeled as IMX415 specifications. New state playback does not silently claim that those old coupled cases were rerun for real hardware.

Optional hardware changes in the source document remain decision options, not implemented approvals. In particular, a white emitter, IR filter change, separate work light or higher-rated connector requires product/hardware decisions and new evidence.

11 Web App and cross-platform handover

Model Lab v2.0 / F2 uses the new W45_N12_G10 per-LED response at five solved distances. It provides manual A/B regimes, all 14 states and a complete sequence, editable physical pulse/fade timings, synchronized playback and 25-bin maps, conditional current/thermal inputs, and the revised requirement/evidence table.

Action	What it computes
Flux, A baseline, solved distance, state / timing	Recomputes fixed-model superposition locally at 0.5ms. No Ansys launch. CV threshold remains blank until measured.
Ambient input	Scenario-only ring/ambient ratio and fraction of combined illuminance; no auto-regime claim.
Playback / matrix export	Play, pause, restart, 0.25–2× viewing speed and loop. Complete CSV includes state, colour policy, slew conflict, current estimate and every 25-bin value at every sample.
PWM / power inputs	Ideal scalar exposure bounds and conditional calibrated-current / one-node steady arithmetic.
Save / JSON import	One F2 browser scenario; version checked; parameters validated and results recomputed. F1 scenario files are not silently accepted.
Geometry / material / spectrum change	New Speos response and numerical verification before use; actual thermal or lens-isolation work needs separate models.

Extract the complete Windows x64 ZIP and double-click Start_Model_Lab_Windows.cmd, or extract the macOS universal ZIP and double-click Start_Model_Lab_macOS.command. Both include official Node runtimes, data and both reports; no Ansys, npm install or Internet is required for local use. Keep the terminal window open; Ctrl+C stops the service. Mac requires 13.5 or newer per the bundled Node platform requirements.

Windows startup, data serving and automated calculations were checked. Mac binaries, archive integrity, executable permissions and launcher syntax were checked; no Mac hardware test was available. Use Copy full contents if the embedded browser does not complete a download; large-file previews are truncated.

12 Missing evidence and delivery register

Test	Evidence still required
M1	CV floor, distance and A/B boundary: Camera/CV metrics, exposure/gain calibration, real emitter output; no prototype available
M2	Material transmission and window uniformity: Measured spectral transmission/scattering, full-window/view protocol; $\pm 20\%$ reference to be agreed
M3	Modulation and auto-exposure: Actual frame/exposure/row timing, AE logs and tolerance
M4	State comfort and fades: Human viewing evaluation and agreed depth/fade criteria
M5	Red-indicator effect on captured frames: RGB spectra, camera spectral response, colour/level tolerances
M6	PWM bands and beats: Part-specific channel/global PWM, clocks, exposure range and image tests
M7	Stray light and image contrast: Lens and front-stack CAD, baffle/BSDF, contrast target and threshold
M8	Current and thermal: Measured current/code curves, standby, burst duration, thermal paths/limits

These are requirement-derived proposed investigation entries, not a reproduction of the missing Investigation brief tab. All physical results remain unfilled. Thresholds for CV, comfort, colour/contrast, real banding and thermal acceptance cannot be established from the available files alone.

Delivery	Location relative to week5_work
Current bilingual report	reports/Harness_Week5_F2_Final_English_Deliverable and Chinese_Reference: PDF / DOCX / MD
Model / requirement evidence	f2/: native .blend, STEP, prepared Workbench, optical exports, transfer arrays, requirements JSON/CSV, workbook and verification records.
Interactive calculator	model-lab/ current F2; model-lab-f1/ preserves the prior source.
Portable packages	launch-packages/Harness_Model_Lab_v2.0_F2_Windows_x64.zip and ...macOS_Universal.zip
Original evidence	The following complete Day1–5 appendices, original F1 reports and source calculations remain historical records.

Priority decisions: confirm window/front-stack CAD and supplier LED data; resolve slew-floor and capture-red conflicts; agree axis/cone and window-uniformity protocols; obtain M1–M8 evidence. Source certainty, simulation assumptions and measured acceptance must remain separate.

A / Day 1: Geometry audit

Historical record. F2 pages 1–12 supersede current requirements, assumptions and delivery status.

01 Geometry baseline and readiness

Monday scope is complete: the Blender and Workbench MCP connections were tested, the supplied lens CAD was audited, an engineering reference was prepared, and an independent minimum Speos optical solve finished successfully. Peter has no physical prototype; the findings below are computational only.

The main correction is geometric: the original cover is a truncated spherical shell with mounting features. Its outer spherical radius of 33 mm does not imply a 66 mm overall enclosure diameter. The earlier Blender hemisphere is unsuitable as the engineering cover.

CAD audit item	Verified result
Units and topology	Millimetres; 1 valid BREP solid, 1 closed shell, 19 faces
Spherical surfaces	Concentric outer R33 / inner R31; 2 mm wall in the spherical region
Top opening diameter	14.1156 mm
Overall X × Y × Z envelope	57.7235 × 58.6000 × 22.3364 mm
Solid volume	7447.7742 mm ³
Engineering frame	Origin at the centre of the lowest mounting reference plane; +Z toward the camera opening

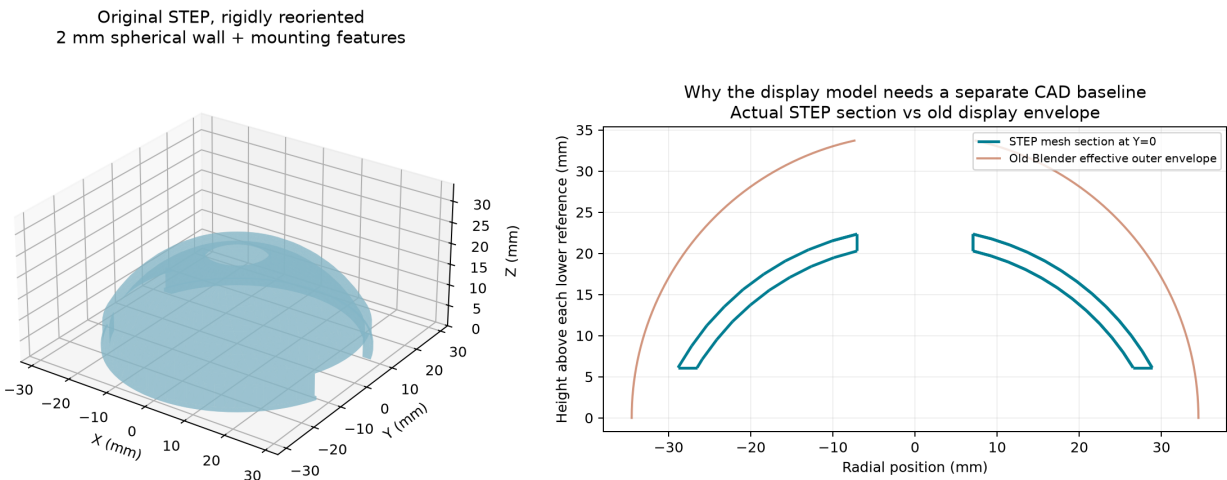


Figure 1. Original CAD geometry and comparison with the earlier Blender approximation.

The normalized STEP uses a rigid coordinate transform and preserves solid validity; the volume difference is below 0.00001 mm³. The Blender display mesh contains 3,312 triangles. Its Y envelope differs from the exact CAD by about 0.039 mm; dimensional authority remains the STEP model.

A separate engineering Blender file preserves the earlier scenes and adds the CAD-derived cover with 12 LED placeholders. The previous 36 animation segments are not a confirmed LED count. Original source files were preserved.

02 Minimum optical solve

An independent Speos 2026 R1 CPU test verified source and sensor creation, ray tracing, result export and numerical readback. This test did not include the device cover or the 12-LED assembly.

Test configuration	Value
Source	2 × 2 mm square Lambertian emitting surface; 1 lm at 555 nm
Receiver	100 × 100 mm plane, 100 mm from the source; 20 × 20 pixels
Ray setting / CPU threads	10,000 rays / 2 threads; engine reported 10,001 emitted rays

Result	Measured or calculated value
Solver state / reported errors	FINISHED / 0
Collected luminous flux	0.241476 lm
Mean receiver illuminance	24.1476 lx
Point-Lambertian reference flux	0.239457 lm
Relative difference from reference	+0.843%



Figure 2. Probe illuminance map. Low ray count causes visible pixel noise.

The reference is a numerical integration of the point-Lambertian approximation. The finite emitting surface and Monte Carlo sampling differ from this approximation. The close integrated-flux agreement is a sanity check, not a convergence result or an acceptance tolerance.

Pixel noise is substantial and some pixels are zero. This run cannot establish local hotspots, product uniformity or camera performance. Native XMP, HTML, TXT and a Speos project export folder were saved.

03 Assumptions, handover and next work

Input status	Meaning for the current model
User supplied	"2 mm LEDs" only; the part number and the meaning of this dimension remain unconfirmed.
Provisional layout	12 emitters; each a 2 × 2 mm square; ring radius 25 mm; height 1.15 mm above the mounting reference plane. These are assumptions.
Unknown physical inputs	LED emission pattern and rated flux; diffuser transmission and scattering; PCB and internal geometry; camera exposure, working distance and power budget.
Normalization for next study	1 lm total for the whole ring, for illuminance comparison in lx/lm. This is not an estimate of actual device output.

Verified software and saved handover

Blender 5.2.2 LTS scene access, code execution and saving were tested. Workbench 2026 R1 was connected and a Speos system was created with the normalized STEP assigned. The saved Workbench project is prepared at geometry level; the complete device model has not been configured or solved. The independent test verifies basic Speos CPU operation only.

- Harness_Week5_Engineering_Reference.blend — CAD-derived cover and provisional LED layout.
- Harness_Week5_Optical.wbpj; geometry/Lens_Engineering_Reference_mm.step — Workbench preparation and exact geometry.
- Parameters_and_Assumptions.csv — 37 parameter records with explicit evidence and assumption categories.
- step_geometry_audit.json; speos_probe_result.json; speos_probe_outputs/optical_sanity_check.json — numerical evidence.

Tuesday: full assembly static baseline

- Build the 12-LED optical assembly using the audited geometry and the stated provisional layout.
- Compare no-cover and ideal transparent-cover controls before exploring diffuser assumptions. Keep ring flux at 1 lm total.
- Report target-plane illuminance per lumen and spatial distributions; check ray-count convergence, geometry discretization and energy balance.
- Replace assumptions as LED, material and camera data become available. Separate conditional simulation comparisons from physical validation.

No claim is made yet about the minimum camera-useful illuminance, real electrical power, thermal limits, final diffuser uniformity, PWM/exposure compatibility or computer-vision success. These require additional inputs and subsequent analysis.

Traceability

Source CAD: 2026-08-20_lens.step. Detailed working record: Day1_Findings.md. All handover filenames above are relative to week5_work. Technical methods: PySpeos launcher and simulation documentation; PyWorkbench existing-session connection documentation. This report records Monday's work only.

B / Day 2: Static optical baseline

Historical record. F2 pages 1–12 supersede current requirements, assumptions and delivery status.

01 Static optical baseline established

The planned Day 2 work was executed on 28 September 2026. A 12-LED model was solved in Ansys Speos 2026 R1 for no-cover and ideal clear-cover controls. Both final cases finished with zero propagation errors. Independent finite-area Lambertian integration and ray-count / mesh refinement checks support the reported whole-region and coarse spatial comparisons.

Model input	Definition and status
Geometry	Audited STEP cover; no LED face intersects it. Minimum face-to-cover clearance: 1.0314 mm.
Emitters — assumed	12 equal 2 × 2 mm square Lambertian faces; ring radius 25 mm; Z=1.15 mm; emitting toward +Z.
Optical normalization	1 lm total (1/12 lm per LED), monochromatic 555 nm. Neither size nor spectrum is a confirmed LED specification.
Clear-cover control	Refractive index 1.5; no bulk absorption or scattering; polished interfaces with Fresnel reflection.
Receiver — assumed	200 × 200 mm planar ROI; distances 0.25, 0.5 and 1 m from the emitter plane; normal -Z.

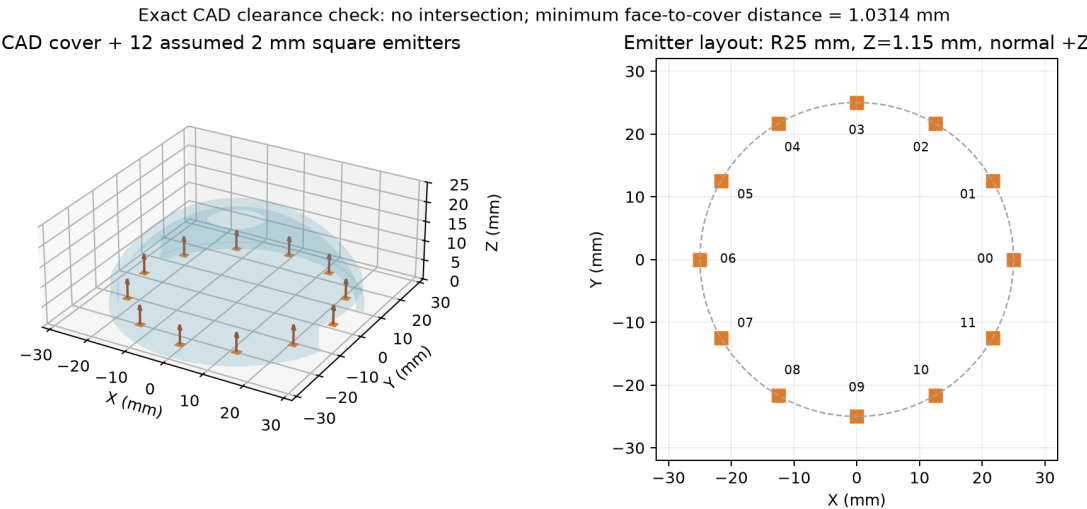


Figure 1. Source placement checked against the original cover CAD; arrows show emission direction.

Distance from emitter plane	Mean lx/lm: no cover → clear cover (change)
0.25 m	4.1462 → 3.9678 (-4.30%)
0.5 m	1.2012 → 1.1967 (-0.37%)
1 m	0.3133 → 0.3161 (+0.90%)

At 0.25 m, the cover changes the captured-region mean by -4.30%. This is a change in regional light collection, not the total optical efficiency of the enclosure. A clear cover does not establish diffuser uniformity or camera usability.

02 Spatial distribution and ray refinement

12-LED baseline | 1 lm total | 32 million rays per run
40 mm spatial bins (5 × 5), common colour scale within each distance

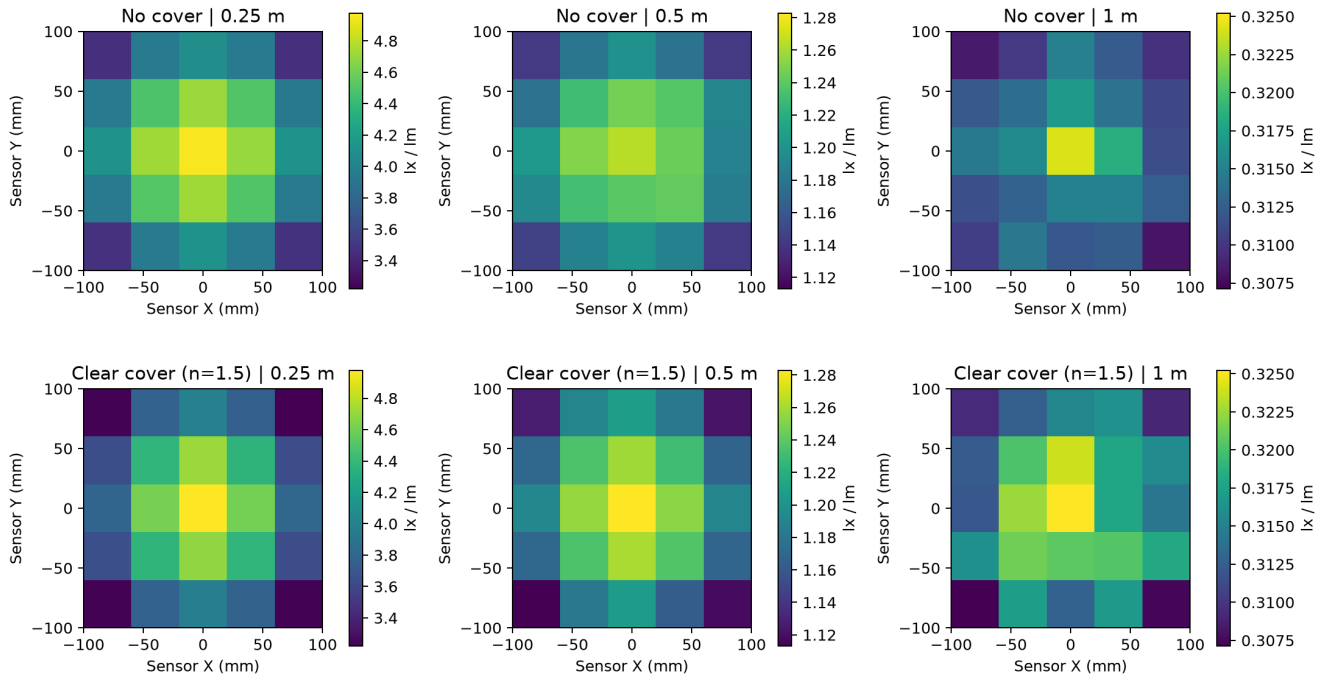


Figure 2. Final 32-million-ray results. Each displayed bin averages 4 × 4 native pixels. The two cases use the same colour scale at each distance.

At 0.25 m, the 5 × 5 bin coefficient of variation rises from 10.39% without the cover to 12.74% with the clear cover. Under this specific model the clear cover does not improve coarse target-plane uniformity. This metric is not cover-surface luminance uniformity.

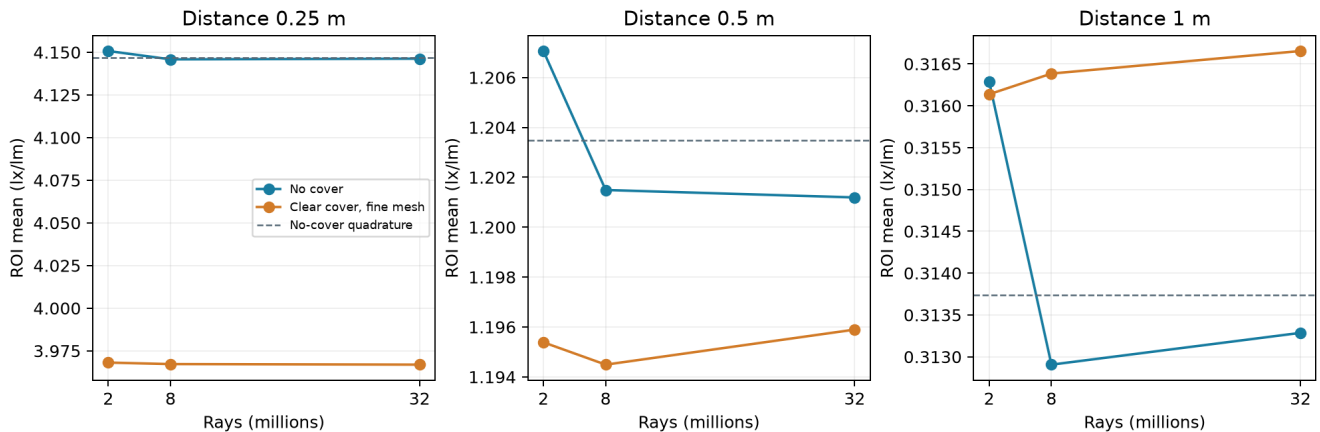


Figure 3. Mean illuminance versus 2, 8 and 32 million rays. The dashed line is the independent finite-area no-cover reference.

The receiver has 20 × 20 native pixels (10 mm pitch). Displayed 5 × 5 bins have 40 mm pitch to reduce sampling noise. Individual native-pixel extrema and small apparent changes at the farther distances should not be treated as resolved hotspots or robust improvements.

03 Numerical verification and handover

Verification	Result
No-cover analytical reference	Maximum mean-illuminance difference: 0.191% across the three distances. Source area and receiver pixel area are both integrated.
Rays: 8 → 32 million	Maximum mean change 0.121%; maximum 5 × 5 map RMS change 1.444% of the refined map mean.
Mesh: 10,790 → 26,460 triangles	Maximum mean change 0.174%; maximum normalized 5 × 5 map RMS change 1.499%. An additional 3,312-triangle case was retained.
Preliminary criteria	Mean change ≤1%; normalized coarse-map RMS change ≤2%; zero propagation errors. Final refinement comparisons meet these criteria; they are numerical screening limits, not product requirements.
Final clear-cover energy balance	Emitted 1 lm; radiated 0.998533 lm; absorbed 0.00146712 lm; error power 0 lm. Rounded report closure residual: 1.2e-07 lm.
Final termination checks	Propagation errors: no cover 0, clear cover 0. Maximum-impact stops: no cover 0, clear cover 0.

The final clear-cover mesh uses flat facet normals, 0.0075 mm sag and 0.05 rad angular settings; geometric ray tolerance is 1e-8 mm and the impact limit is 2,000. The earlier smooth-normal meshes produced propagation errors and are diagnostic records only. Ray tolerance is a solver setting, not manufacturing accuracy. A 38,710-triangle assembly exceeded the Student licence limit of fewer than 32,000 triangles; the final assembly has 26,484 triangles.

Model limits and next decision

- The cover is lossless in bulk; absorption in the energy report occurs at the ideal black LED surfaces. PCB, opaque housing, camera and actual LED packages are absent. Returned light can escape through the open underside.
- Real LED part, spectrum and emission data, diffuser scattering, internal geometry and camera task requirements remain unknown. There is no electrical-power, thermal, diffuser-hotspot or CV-performance conclusion.
- Next: compare explicitly hypothetical scattering conditions and layout changes while holding total ring flux fixed. Add cover-surface and camera-opening diagnostics before selecting a candidate.

Saved evidence and reproduction

day2/baseline_metrics.csv and comparison_summary.json contain the results and refinement checks. Each final run under day2/runs contains XMP, TXT, HTML, NumPy maps and a native Speos export directory. day2/README.md identifies the accepted runs and reproduction commands. Original Workbench and Blender projects were preserved; the optical controls were solved through the Speos RPC API.

C / Day 3: Material/layout sensitivity

Historical record. F2 pages 1–12 supersede current requirements, assumptions and delivery status.

01 Material and layout sensitivity study

The planned Day 3 study was executed on 28 September 2026 using Speos 2026 R1. Fourteen hypothetical cases compare bulk scattering, LED count, placement and absorption. The objective is to expose trade-offs and retain candidates for better input data; no actual diffuser material or final hardware specification is selected.

Common inputs: audited cover STEP, 2 × 2 mm Lambertian emitting faces, 1 lm total ring flux and 555 nm. Default layout: 12 LEDs at R25 mm, Z1.15 mm. Material index is 1.5, interfaces polished, absorption zero unless stated. Targets remain fixed at Z251.15, 501.15 and 1001.15 mm over a 200 × 200 mm region, including when LED height changes.

Henyey–Greenstein bulk scattering is parameterized by coefficient μ_s and anisotropy g . The μ_s values 0.25, 1 and 3 /mm give 2 mm normal-path optical depths 0.5, 2 and 6. These are exploratory levels, not measured polymer properties or a surface-frosting specification. Unless changed, $g=0.7$.

Case	Change from reference layout/material
C0_clear	Clear control; scattering = 0
S025	Scattering coefficient 0.25 /mm
S100	Reference: scattering 1 /mm, $g=0.7$
S300	Scattering coefficient 3 /mm
G000	S100 with $g=0$ (isotropic scattering)
G090	S100 with $g=0.9$ (more forward scattering)
N16	S100 with 16 LEDs
N24	S100 with 24 LEDs
R22	S100 with ring radius 22 mm
R26	S100 with ring radius 26 mm; clearance risk
Z415	S100 with emitting-plane height 4.15 mm
A002	S100 with absorption coefficient 0.02 /mm
N24_S050	24 LEDs + scattering 0.5 /mm
N24_R22	24 LEDs + radius 22 mm, scattering 1 /mm

All emitting faces clear the CAD cover. R26 leaves only 0.032 mm clearance and is retained as a numerical boundary case, not a recommended assembly. The real LED package is absent from every clearance calculation.

02 Quantified trade-offs

Case	E at 0.25 m (lx/lm)	CV ring / full cover	Opening proxy (% flux)	Rays (M)
C0_clear	3.972	2.616 / 3.044	2.353	8
S025	3.135	2.094 / 2.269	2.470	8
S100	2.011	0.816 / 0.826	2.680	8
S300	1.354	0.219 / 0.312	2.921	16
G000	1.473	0.979 / 0.931	2.923	8
G090	2.718	1.075 / 1.186	2.574	8
N16	2.026	0.706 / 0.733	2.670	8
N24	2.037	0.597 / 0.801	2.674	8
R22	2.209	0.544 / 0.604	3.053	8
R26	1.844	0.845 / 0.955	3.033	8
Z415	2.146	1.125 / 1.185	2.958	8
A002	1.760	0.885 / 0.891	2.589	8
N24_S050	2.648	1.031 / 1.162	2.557	16
N24_R22	2.222	0.357 / 0.450	3.042	16

E is target-plane mean illuminance per lumen of total ring flux. CV is luminance standard deviation divided by mean, shown for the ring and full projected cover; lower is smoother. Opening proxy is upward light crossing a circular monitor below the opening, not camera-sensor exposure or glare. Values at 0.5 and 1 m are supplied in study_results.csv.

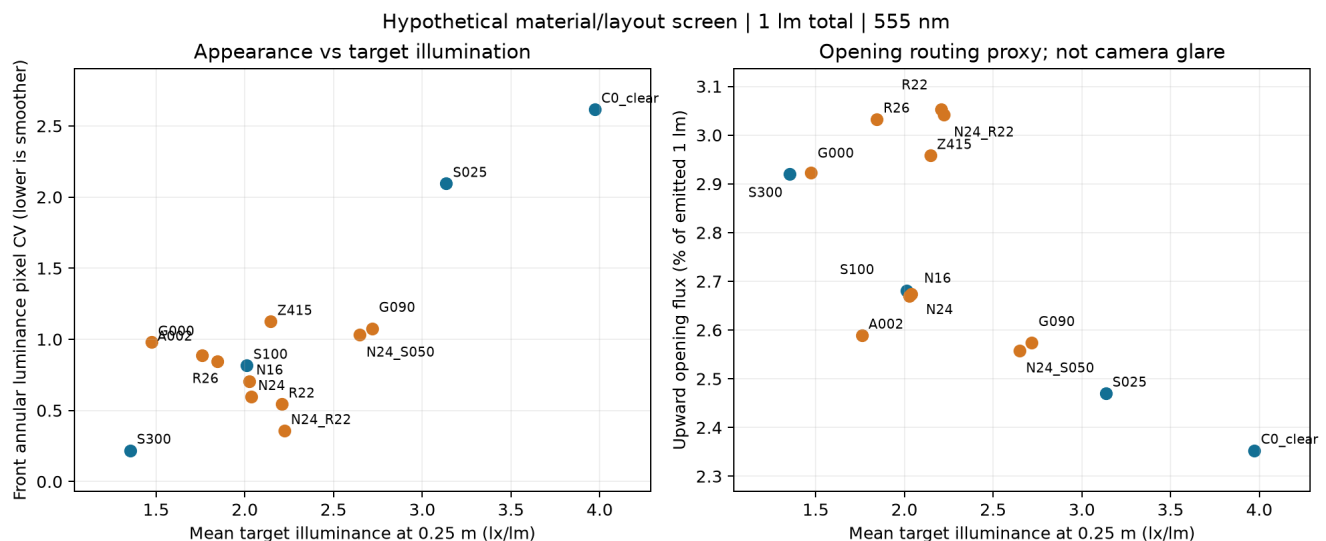


Figure 1. Higher illumination and lower luminance CV are competing objectives; opening routing adds a separate constraint.

For the 12-LED reference, S100 loses 49.4% of E250 relative to clear, while S300 loses 65.9%. Scattering improves the sampled appearance, but improved appearance alone is insufficient to choose a design.

03 Appearance comparison at a fixed view

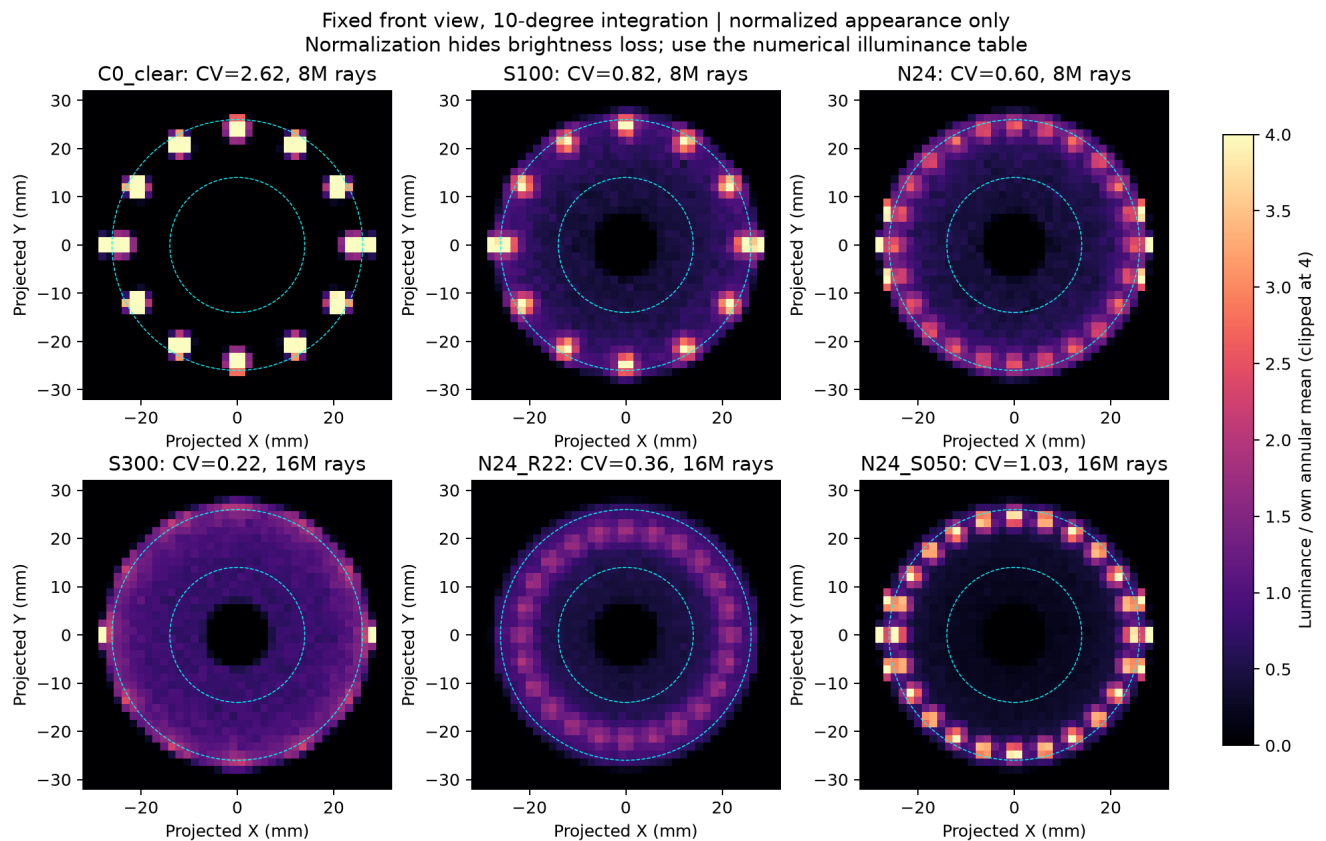


Figure 2. Each image is divided by its own annular mean, so these images deliberately suppress absolute brightness differences. The cyan rings bound the measurement region; values above 4 are clipped.

The photometric radiance sensor uses a 64×64 mm image plane at Z24 mm, a front observer at Z524 mm, 40×40 pixels and a fixed 10-degree integration angle. Metrics use pixel centres at projected radius 14–26 mm. This is a finite-cone directional luminance diagnostic; it does not establish all-view appearance or perceptual comfort.

With the same S100 material and total flux, increasing from 12 to 24 LEDs changes the sampled front CV from 0.816 to 0.597, and E250 from 2.011 to 2.037 lx/lm. The added emitting area and changed spatial sampling are part of this layout change; per-LED output is reduced accordingly.

An additional mask of 892 pixels, obtained by exact CAD ray intersections from the central observer, includes the mounting edges and near-hole region. Strong scattering still leaves bright edge regions. Pixel CV includes radial gradients as well as individual bright spots. Native pixels are 1.6 mm on the image plane. Angular-sector summaries can hide periodic LED patterns if bins align with spot boundaries; the full fixed-mask pixel CV is therefore the primary metric. Small differences near numerical noise are not used to rank candidates.

The opening monitor is at Z20 mm with radius 7.0578 mm. Its 40×40 map is integrated with fractional circular pixel weights. Some monitored rays can interact with the opening walls later; this records routing toward the opening footprint, not confirmed transmission through a real camera entrance pupil.

04 Candidate directions and verification

Candidate	Reason to retain and cost
N24_S050 / Brightness-leaning	E250=2.648 lx/lm; CV ring/full=1.031/1.162. Relative to clear: 33.3% less target illuminance. Opening proxy 2.557%. 24 LEDs; clearance 0.986 mm.
N24_R22 / Balanced optical direction	E250=2.222 lx/lm; CV ring/full=0.357/0.450. Relative to clear: 44.0% less target illuminance. Opening proxy 3.042%. 24 LEDs; clearance 3.986 mm.
S300 / Smoothness-leaning	E250=1.354 lx/lm; CV ring/full=0.219/0.312. Relative to clear: 65.9% less target illuminance. Opening proxy 2.921%. 12 LEDs; clearance 1.031 mm.

These are conditional alternatives. G090 achieves similar performance to N24_S050 with 12 LEDs; the small contrast difference does not justify 24 LEDs by itself. Selection requires an illumination floor, acceptable contrast and actual LED/material data. Normalized lumen needs do not directly establish electrical power.

Numerical check	Evidence
Radiance sensor calibration	Known 20 mm Lambertian square: expected 795.775 cd/m ² ; measured 794.968 cd/m ² (-0.10%).
Candidate ray refinement	8M rays compared with higher-count runs: maximum scalar mean/opening change 0.72%; maximum absolute front-CV change 0.017. Screening limits: 2% and 0.03 respectively.
Geometry / tracing	26,460 cover facets; total geometry below 32,000 Student limit. Flat facet normals; 1e-8 mm ray tolerance; max 2,000 impacts. All reported cases have zero propagation errors.
Optical energy balance	Source 1 lm = absorbed + radiated + error/stopped components to HTML rounding. Target and opening monitors are not summed as separate emitted power.

The bulk material format follows Ansys documentation; coefficients are deliberately hypothetical. The study omits PCB, opaque housing, camera optics and actual LED packages. No physical sample, broadband colour behaviour, thermal result, PWM compatibility or CV success was verified. Mesh validation from Day 2 supports this mesh family; full new-material spatial and angular convergence remains beyond this screen.

Handover: day3/study_results.csv, study_summary.json, study_manifest.json, materials/, accepted runs/ and sensor_calibration/. Each run retains native XMP, TXT, HTML and Speos exports. Reproduction and the exact candidate selection rationale are in day3/README.md. Next: obtain material/LED data and define the illumination floor before dynamic-state and power analysis.

D / Day 4: Dynamics and exposure

Historical record. F2 pages 1–12 supersede current requirements, assumptions and delivery status.

01 Dynamic illumination model

Day 4 work applies the existing four-state animation engine to the conditional N24_R22 optical candidate. Twenty-four separate Speos solves measure each LED’s target response. Their sum is checked against the prior equal-output result and a new unequal-output direct solve. The camera and driver remain unspecified; this report establishes conditional relationships, not hardware acceptance.

Common optical inputs	Model
Layout / cover	24 provisional 2 × 2 mm emitters; R22 mm, Z1.15 mm; audited CAD cover.
Material / source	Hypothetical bulk scattering 1/mm, g=0.7, index 1.5, zero bulk absorption; polished interfaces; Lambertian emission at 555 nm.
Targets / resolution	200 × 200 mm targets at 0.25, 0.5 and 1 m from the baseline emitter plane. Native 20 × 20 maps are averaged into 5 × 5 bins, each 40 mm square.
Transfer coefficient	$K_{[j,i]}$ is illuminance in target bin j per lumen emitted by LED i . Every LED face remains in each scene, including inactive black faces.

Define Φ_{peak} as total source flux when all 24 LEDs have command 1. With command a_i , LED i emits $\Phi_{\text{peak}} \times a_i / 24$. Then $E_j(t) = \Phi_{\text{peak}} \times \sum_i K_{[j,i]} a_i(t) / 24$. Changing the spectrum, material, geometry or emission pattern requires a new optical model.

A test baseline without changing the application

The original Week 4 engine is imported directly, with only LED count changed to 24. Its global brightness 0.55, 2.8 s Listening cycle, 4 s / 90° Processing arc, opposed Speaking B lobes and Error B signature are retained. Ordinary transitions remain 0.5 s. No application files were modified.

Compare the original command s_i with $a_i = 0.6 + 0.4 s_i$. The 60% value is an exploratory baseline; 20%, 40% and 80% are also scanned. This raises average flux and reduces state contrast, so it is not a matched-power appearance comparison. Recognition and comfort were not tested.

Numerical check	Evidence
Optical superposition	2M rays per LED; unequal-output check 8M rays. Largest target mean difference 0.73%; map RMS / reference mean 2.17%. Preset limits: 2% and 4%, plus 12% maximum-bin difference / mean.
Temporal refinement	0.5 ms vs 1 ms sampling: largest minimum-illuminance change 0.012%; preset limit 0.2%. All optical runs have zero propagation errors and max-impact stops.

02 State illumination floors

State	Existing min	With baseline	Average flux fraction
Listening	0.0595	0.3845	0.708
Processing	0.0619	0.3859	0.648
Speaking	0.0538	0.3822	0.646
Error	0.0510	0.3811	0.641

Values are lowest 40 mm target-bin illuminance at 0.5 m, in lx per lumen of Φ_{peak} . The last column is emitted flux averaged over the state, divided by Φ_{peak} . The baseline raises average source flux by 2.6–6.2 times versus the original commands at the same scale. Speaking covers a 6.4 s synthetic sample only.

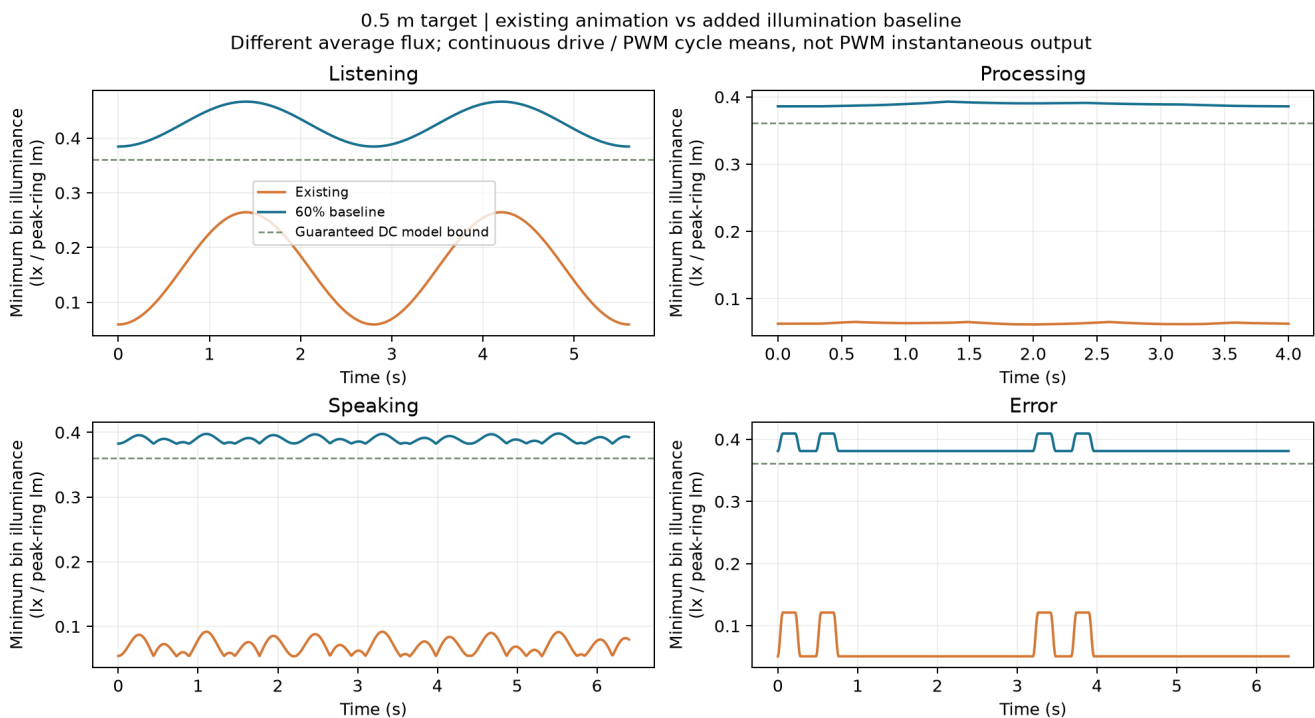


Figure 1. Original state commands versus the added 60% baseline; both use the same peak-ring flux scale.

For a linear, nonnegative transfer model, $a_i \geq \beta$ implies $E_j \geq \beta \times E_{\text{all-on},j}$ at every time. This bound is valid for continuous amplitude drive. If the entire output uses aligned PWM, it bounds the intended cycle mean only: every LED can still be off during a carrier interval.

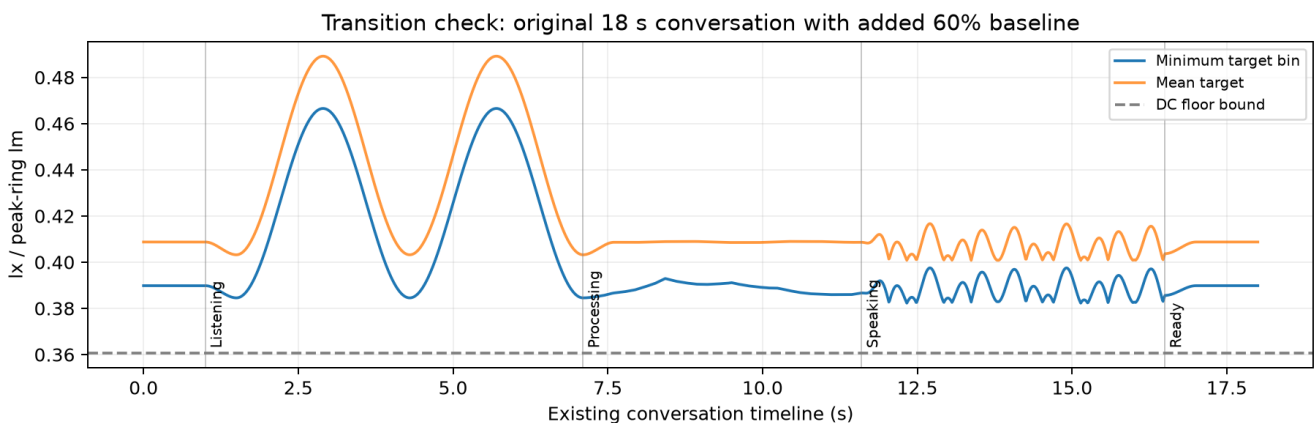


Figure 2. The original 18 s conversation, including transitions, retains the added model baseline. No user-recognition result is implied.

03 PWM and exposure: evaluate timing together

For ideal constant duty D and exposure T , let $fT = n + r$, where n is an integer and $0 \leq r < 1$. The lowest exposure-mean output over all phases is $[nD + \max(0, r - 1 + D)] / (fT)$; the highest is $[nD + \min(r, D)] / (fT)$. Compare these with expected mean D .

PWM / exposure	Duty	Worst phase error
1 kHz / 0.2 ms	60%	100.00%
2 kHz / 0.37 ms	60%	35.14%
20 kHz / 0.2 ms	60%	0.00%
20 kHz / 0.37 ms	60%	5.41%

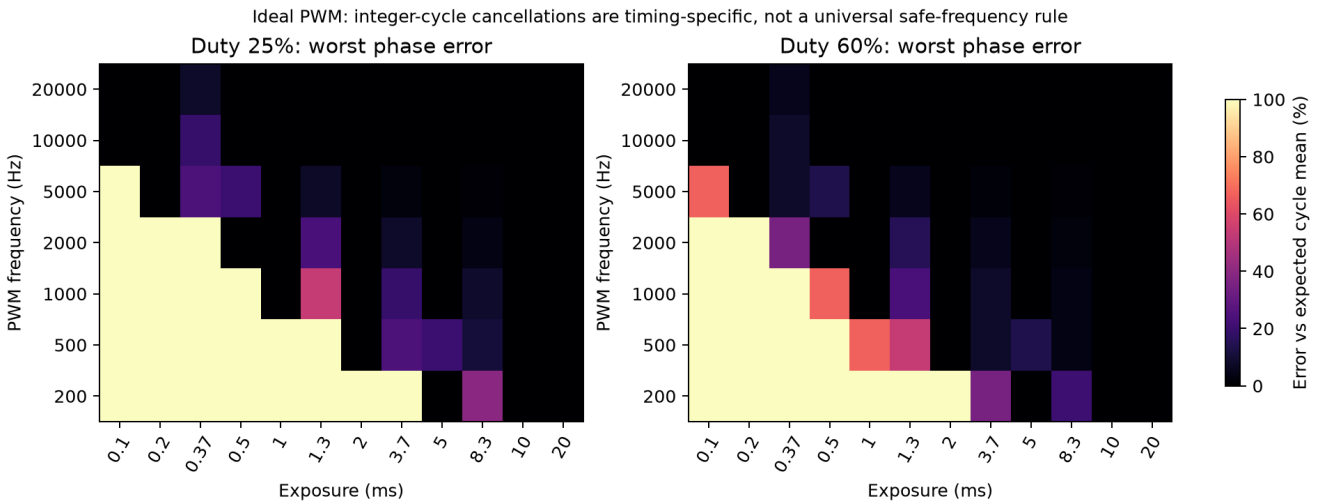


Figure 3. Exact phase extremes for ideal constant-duty PWM. Error is relative to the expected cycle mean; values above 100% are clipped in the colour scale.

An exposure spanning an integer number of ideal carrier cycles cancels constant-duty phase error. This is a timing coincidence, not proof of a safe frequency. Non-integer exposure cases and a nine-point $\pm 1\%$ carrier-frequency scan are included in the CSV. Duty changes, clock error and real switching can spoil cancellation.

A conservative phase-independent relative-error bound is $(1 - D)/(fT)$. It helps express a requirement once minimum exposure and duty are known; it does not select a driver by itself. The analytic results were checked at all piecewise breakpoints and against an independent numerical pulse integral.

Basler documents short-exposure PWM brightness fluctuations and rolling-shutter stripes for its camera lights. Its camera-specific example is not adopted as a universal Harness threshold. Method context: docs.baslerweb.com/light-control and [/electronic-shutter-types](https://docs.baslerweb.com/electronic-shutter-types); links are preserved in [day4/README.md](#).

04 Coupled state and rolling-row exposure

The existing 18 s conversation and 6.4 s Error waveform are integrated with ideal per-LED PWM, with duty latched at each carrier edge. Exploratory timing: 30 fps, 480 virtual rows, 20 μ s row offset, 16 frame/carrier phase offsets, exposures 0.2 / 1.3 / 5 ms and carriers 1 / 2 / 20 kHz. These values are scenarios, not specifications of Peter's camera.

Carrier / drive	Global error	Row error	Row contrast
1 kHz / full PWM	100.00%	100.00%	157.59%
2 kHz / full PWM	85.76%	86.38%	143.97%
20 kHz / full PWM	0.01%	0.01%	0.01%
2 kHz / DC + PWM	14.93%	15.20%	27.26%

Table: Error, 0.2 ms exposure. Errors compare exposure-integrated PWM illumination with the continuous-command reference over the identical exposure. Row contrast is the within-frame range of that ratio. Every virtual row views the same static central target patch at 0.5 m, isolating temporal stripe risk from scene shading.

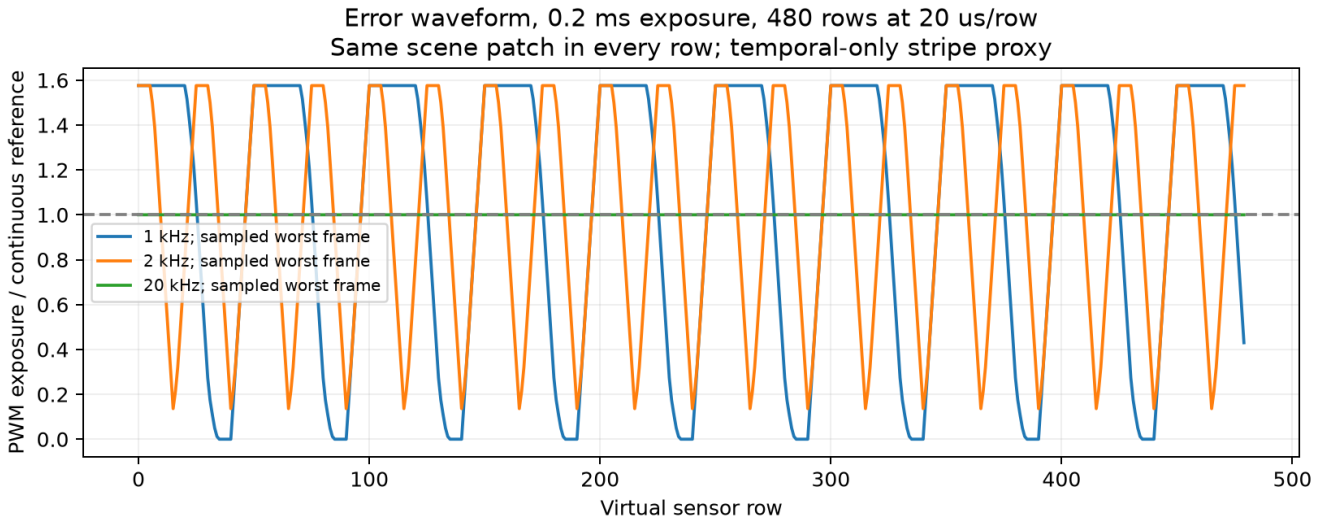


Figure 4. Sampled worst Error frames for each carrier. The vertical coordinate is a temporal illumination ratio, not simulated sensor pixel brightness.

Full PWM gates the entire command. The hypothetical hybrid case keeps a 60% continuously lit component and PWM-controls only the remaining overlay. The hybrid has a true model illumination floor even between PWM pulses, but requires a suitable driver and verified flux-versus-current behaviour; implementation is not established.

Higher carrier frequency reduces errors in many sampled cases, but this finite phase/frame grid is not a worst-case guarantee for all cameras. No optics, scene reflectance, autoexposure, sensor noise, motion, colour processing or CV task is modeled. Privacy/capture overrides and Sleep are outside this illumination-required scope.

05 Conditional operating envelope and handover

Example only: require every 40 mm target bin to receive at least 5 lx. The first scale below covers the sampled state windows. The larger DC-bound scale uses $\Phi_{\text{peak}} = E_{\text{req}} / [0.6 \times \min(E_{\text{all-on}})]$ and covers any command waveform that preserves the continuous baseline in this optical model. The 5 lx level is not a validated camera requirement.

Distance	Sampled-state scale (lm)	DC-bound scale (lm)	Per LED at DC-bound scale (lm)
0.25 m	4.1	4.4	0.18
0.5 m	13.1	13.9	0.58
1 m	49.1	52.0	2.17

All values scale linearly with E_{req} under the fixed 555 nm model. The per-LED value is the command-1 flux scale, not a claim that a 2 mm LED can supply it. Actual output, spectrum, packaging, duty limits and thermal droop must be checked before translating these values into a component choice.

Electrical / thermal relationship	What is still unknown
$\text{PLED}(t) = \Phi_{\text{emitted}}(t) / \eta_{\text{LED}}$	LED electrical luminous efficacy η_{LED} in lm/W at the modeled spectrum and operating point.
$\text{Pin}_{\text{put}} = \text{PLED} / \eta_{\text{driver}}$	Driver efficiency; driver loss = $\text{Pin}_{\text{put}} - \text{PLED}$.
Conservative steady thermal screen: $\Delta T = R_{\theta, \text{eff}} \times \text{mean}(\text{Pin}_{\text{put}})$	Assumes all input becomes heat at the chosen thermal node. Effective thermal resistance, ambient temperature, allowable rise and component hot spots are unknown. This is not an Ansys thermal solve.

The CSV reports source-flux coefficients so an actual efficacy and driver efficiency can be substituted later. Time-average power controls the simplified steady thermal estimate; instantaneous LED/driver limits must separately accommodate peaks. Do not substitute a junction-to-case resistance for whole-device thermal resistance.

Requirements supported by this calculation

R1/R2: normalized target illumination and a conditional floor. R3/R4: existing timing is evaluated, but recognition and comfort remain untested. R6: explicit PWM/exposure risk scenarios, not camera compatibility approval. R7: power and thermal equations only. Colour-dependent optics, source-beam sensitivity and internal housing/PCB occlusion remain open.

Handover: day4/runs/, transfer_lx_per_LED_lm.npy, waveforms/, dynamic_results.csv, baseline_sweep.csv, conditional_operating_envelope.csv, pwm_phase_scan.csv and coupled_exposure_results.csv. Verification JSON and README provide input provenance and reproduction steps. Next: consolidate R1–R7 into measurable draft requirements and a data-request / prototype-validation matrix.

E / Day 5: Original R1–R7 draft

Historical record. F2 pages 1–12 supersede current requirements, assumptions and delivery status.

01 Week 5 outcome and draft status

Revision v0.1, prepared 28 September 2026. Peter’s calculation and documentation package is complete for the currently available inputs. The planned Day 1–5 sequence was executed early; no physical prototype was tested. Source: Harness Lighting System Week 5: Preliminary Requirements, 26 September 2026.

The working result is a reproducible optical and temporal model, three conditional diffuser/layout directions, and measurable R1–R7 draft requirements. No requirement is marked as physically passed. Source-beam sensitivity, real spectra/materials, internal occlusion and numerical thermal estimates remain open.

ID	Current evidence / open acceptance
R1 Scene illumination	Normalized optical response available; camera threshold and real scene unverified.
R2 Illumination during state changes	Continuous-drive model bound demonstrated; full aligned PWM can still go dark.
R3 Device-state communication	Four state families implemented; recognition not measured.
R4 Non-jarring behaviour	Timing and command slopes recorded; perceptual limits unverified.
R5 Diffused output	Fourteen hypothetical cases screened; material, other views and final thresholds unknown.
R6 Camera compatibility	PWM/exposure risks calculated; no real-camera compatibility or glare result.
R7 Power and thermal limits	Conditional equations only; no component selection, wattage approval or thermal solve.

Evidence that can be carried forward

Audited CAD replaces the earlier hemisphere approximation. Day 3 screened 14 hypothetical cases; Day 4 used 24 single-LED responses and a new unequal-output direct solve. Superposition differences were at most 0.73% in target means and 2.17% in map RMS/reference mean. These are numerical checks, not overall physical-accuracy bounds.

Common assumptions: 2 × 2 mm emitting faces for the user’s unspecified “2 mm LEDs”; 555 nm Lambertian emission; hypothetical diffuser optics; no PCB, opaque housing or real camera. The 0.25 / 0.5 / 1 m distances and 200 mm target square are study cases, not a confirmed working range. All acceptance parameters remain TBD in the register.

R1 Scene illumination

Across the agreed operating scene, target ROI and working distances, the lighting shall provide minimum target-bin illuminance $E_{\min} \geq E_{\text{req}}$ and meet the agreed camera/CV task criterion. Nominal and maximum output shall be defined separately.

Report $E_{\min}$, E_{mean} and $E_{\min}/E_{\text{mean}}$ at a fixed spatial sampling scale, ambient level, target reflectance and camera settings; compare task score against A_{task} . Lux alone does not establish CV success.

Traceability: EV2, EV3, EV4 / P01, P02, P03, P04 / V01. Normalized optical response available; camera threshold and real scene unverified.

R2 Illumination during state changes

Whenever scene illumination is required, every state and transition shall preserve a continuous target illumination floor E_{floor} ; exposure-integrated light shall also remain above Q_{floor} for supported camera timing. The time basis of the floor must be explicitly agreed.

Measure min over target bins and time of $E_j(t)$; measure $Q_j = \int E_j dt$ over each exposure. Keep instantaneous, PWM-cycle mean and exposure-integrated criteria separate. An exposure-only interpretation requires an explicit requirement revision.

Traceability: EV4, EV5 / P05, P06 / V02, V05. Continuous-drive model bound demonstrated; full aligned PWM can still go dark.

Conditional scale example — not a camera specification

Distance	$E_{\text{all-on,min}}$ (lx/lm)	Φ_{peak} for 5 lx (lm)	Per LED at command 1 (lm)
0.25 m	1.9137	4.4	0.18
0.5 m	0.6012	13.9	0.58
1 m	0.1603	52.0	2.17

N24_R22; $\beta=0.6$ continuous baseline; 40 mm target bins. $\Phi_{\text{peak}} = E_{\text{req}}/[\beta \times \min(E_{\text{all-on}})]$ is the source flux scale if all 24 LEDs were at command 1. The table uses $E_{\text{req}}=5$ lx only as an illustration, without manufacturing/ageing margin. Actual state-average flux is lower and depends on the waveform.

Neither 5 lx, 60%, 24 LEDs nor the listed per-LED lumens is an approved specification. Real spectra/beam shapes require recalculation. Camera-task data determine E_{req} ; LED/driver data determine whether the output is feasible. R2 retains a continuous-floor draft: a PWM cycle mean must not silently replace it.

R3 Device-state communication

Users in the agreed viewing conditions shall distinguish Listening, Processing, Speaking and Error with recognition rate $\geq A_{rec}$ and response time $\leq t_{rec}$; confusion between state pairs shall be reported.

Use a balanced, randomized, unlabeled recognition task; predefine participant count N_{user} , trials, viewing positions and decision rule. Existing visual-engine behaviour is implementation evidence, not recognition evidence.

Traceability: EV6, EV4 / P07, P08 / V03. Four state families implemented; recognition not measured.

R4 Non-jarring behaviour

Normal state changes shall stay within agreed transition-duration, visible modulation and luminance-rate limits and pass a predefined dark-room discomfort criterion. Error accents require their own stated limits.

Measure transition time, $m=(L_{max}-L_{min})/(L_{max}+L_{min})$, calibrated dL/dt and predefined discomfort score/fraction. Do not equate normalized command slope with physical luminance or comfort.

Traceability: EV6, EV4 / P09, P10 / V02, V03. Timing and command slopes recorded; perceptual limits unverified.

Inherited design signal	Value / limit status
Listening	2.8 s full-ring breath
Processing	90° moving arc; 4 s per turn
Speaking B	Opposed patches; synthetic envelope; strength 0.55
Error B	Persistent three marks; 3.2 s repetition; 0.28 s pulses at 0 and 0.48 s; 60 ms eased edges
Ordinary transitions	0.5 s; recognition and comfort not verified
Day 4 baseline	$a = 0.6 + 0.4s$; exploratory, not production firmware

The state engine was sampled at 24 LEDs with its existing global brightness 0.55. The added 60% baseline increases average emitted flux by about 2.6–6.2 times at the same peak scale and reduces relative state contrast. Its illumination advantage does not establish recognizability, preference or comfort.

V03 separates an unlabeled recognition task from preference/comfort ratings. Participant count, viewing conditions, trial allocation and decision rule must be set before testing. Capture/privacy overrides, wake and sleep were excluded from the Day 4 illumination-required study and need an explicit policy before closing requirements.

R5 Diffused output

At the agreed observation positions, output levels and calibrated imaging settings, the complete visible cover shall meet $CV_L \leq U_{CV}$ and hotspot ratio $L95/L_{mean} \leq H95$; local mounting-edge or aperture hotspots shall be inspected separately.

Fix masks, pixel scale, observer geometry and angular integration. Report full-cover and annular results separately, plus absolute luminance. A normalized rendering or a single scalar can hide local bright spots.

Traceability: EV3 / P11, P12 / V04. Fourteen hypothetical cases screened; material, other views and final thresholds unknown.

Case	E250 (lx/lm)	CV ring / full	Opening proxy
N24_S050	2.648	1.031 / 1.162	2.557%
N24_R22	2.222	0.357 / 0.450	3.042%
S300	1.354	0.219 / 0.312	2.921%

Values use 1 lm total at 555 nm. CV is a fixed-front-view pixel statistic; lower is smoother only within that measurement setup. Stronger scattering reduces E250. Retain N24_R22 as a next-model reference, not a selected part or production LED count. The opening number is upward routing through a monitor below the opening, not actual camera throughput.

R6 Camera compatibility

Throughout the supported exposure, shutter, frame-rate and lighting range, exposure error, row variation and stray-light degradation shall remain below agreed limits; the camera/CV task shall meet its acceptance criterion.

Temporal metrics: $\epsilon_Q = \max|Q_{PWM}/Q_{DC} - 1|$ and $B_{row} = \max(\text{row ratios}) - \min(\text{row ratios})$, with the same scene and exposure reference. For stray light, define a calibrated raw dark-patch/contrast degradation metric G_{veil} and its reference. Opening-plane flux is a routing proxy, not a camera pass/fail metric.

Traceability: EV3, EV5 / P13, P14, P15 / V05. PWM/exposure risks calculated; no real-camera compatibility or glare result.

The ideal 60%-duty calculation gives a 100% worst-phase exposure deficit at 1 kHz / 0.2 ms. At 20 kHz / 0.2 ms ideal constant-duty phase error cancels, but 20 kHz / 0.37 ms still gives 5.41% worst error. These illustrate timing dependence; no frequency is approved. Actual camera optics and G_{veil} remain unmodeled.

R7 Power and thermal limits

For all supported lighting states, transitions and ambient conditions, LED and driver peak/average currents and powers shall remain within device ratings and the system budget; specified component and accessible-surface temperatures shall remain within approved limits for the required duration.

Record input voltage/current waveforms, LED/driver dissipation, temperatures and flux droop. Define duty cycle, run duration and thermal equilibrium criterion before test. Separate source lumens, emitted radiant watts and electrical watts.

Traceability: EV4 / P16, P17, P18 / V06. Conditional equations only; no component selection, wattage approval or thermal solve.

Conditional budget: $P_{LED}(t) = \Phi_{emitted}(t) / \eta_{LED}$; $P_{input} = P_{LED} / \eta_{driver}$. A single-node conservative heat-input screen uses $\Delta T = R_{\theta, eff} \times \text{mean}(P_{input})$, treating all electrical input as heat at that node. It is not an all-hotspot guarantee. η_{LED} , driver efficiency, heat paths, ambient and limits are unknown; no numeric wattage or temperature is approved.

Data requests — proposed roles, not assigned commitments

Input / priority	Suggested provider and missing data
D01 / P0	EE/supplier: LED part, flux/spectrum, angular data and ratings
D02 / P0	Optics/supplier: measured diffuser properties and tolerances
D03 / P0	Mechanical/optics: PCB, housing, package, camera and baffle CAD
D04 / P0	Camera/CV: shutter, exposure, frame/row timing, control policy
D05 / P0	System/CV: working scene, ROI and task acceptance metric
D06 / P0	EE/firmware: driver architecture, electrical budgets and timing
D07 / P0	Thermal/mechanical: heat paths, ambient and temperature limits
D08 / P1	UX/product: recognition, comfort, views and hotspot criteria

P0 data gate numerical hardware decisions; P1 still gates perception-related acceptance. Peter or the project lead can coordinate collection. No requests have been sent externally. Data_Requests.csv contains the exact requested fields and what each group unlocks.

06 Verification plan and handover

Plan / requirements	Future measurement and decision
V01 / R1	Calibrated target illumination plus camera task score over agreed scenes; set E_req before acceptance.
V02 / R2,R4	Resolve carrier pulses with suitable photodetection; log instantaneous floor, cycle mean and exposure integral separately.
V03 / R3,R4	Randomized unlabeled recognition plus separately scored comfort; predeclare participants, trials and criteria.
V04 / R5	Calibrated full-cover luminance at agreed views; include edges and aperture, absolute levels and normalized metrics.
V05 / R2,R6	Actual camera/driver timing sweep with matched continuous-light reference; test stray light with real camera/baffle.
V06 / R7	Measure input/channel electrical peaks and averages; log temperature, droop and equilibrium at specified ambient/duration.

All V01–V06 tests are planned, not executed. Record instruments/calibration, exact assembly/firmware, scene and environment, raw data, uncertainty and failure cases. The blank test register intentionally contains no operator, date, measured value or pass/fail result. Set thresholds and the decision rule before measurement.

Order of the next iteration

- Collect D01–D07 and agree scene/task/floor definitions. Resolve what “2 mm” describes; identify the camera and driver.
- Replace proxies and add housing, PCB, package and camera/baffle geometry. Recompute changed optical responses; check actual electrical feasibility and thermal inputs.
- Agree perception criteria with D08, then execute V01–V06 when hardware and participants are available. Revise requirement values against evidence before design release.

Editable deliverables and evidence navigation

day5/ contains the EN/ZH requirements CSVs, Open_Parameters.csv, Data_Requests.csv, Prototype_Validation_Plan.csv and a blank test-record CSV. Requirements_Review.xlsx combines these as review sheets. requirements_package.json links each requirement to EV evidence groups, P parameters, D inputs and V tests, with evidence-file hashes.

Detailed computation remains in day2/, day3/ and day4/ and their bilingual reports. The main entry point is Week5_Handover_Index.md. Numerical solver tolerances are not product limits. The source document is treated as requirements background; this draft follows Peter’s no-hardware, calculation-and-simulation scope. Nothing in this handover records physical sign-off.